

Advanced Imaging Technologies in Systems Biology and Biomedical Research

Periklis (Laki) Pantazis

What we covered

- How to visualise fluorescent probes in thick specimen?
 - How to achieve deep tissue imaging?
-

Confocal optical section of a specimen - the principle

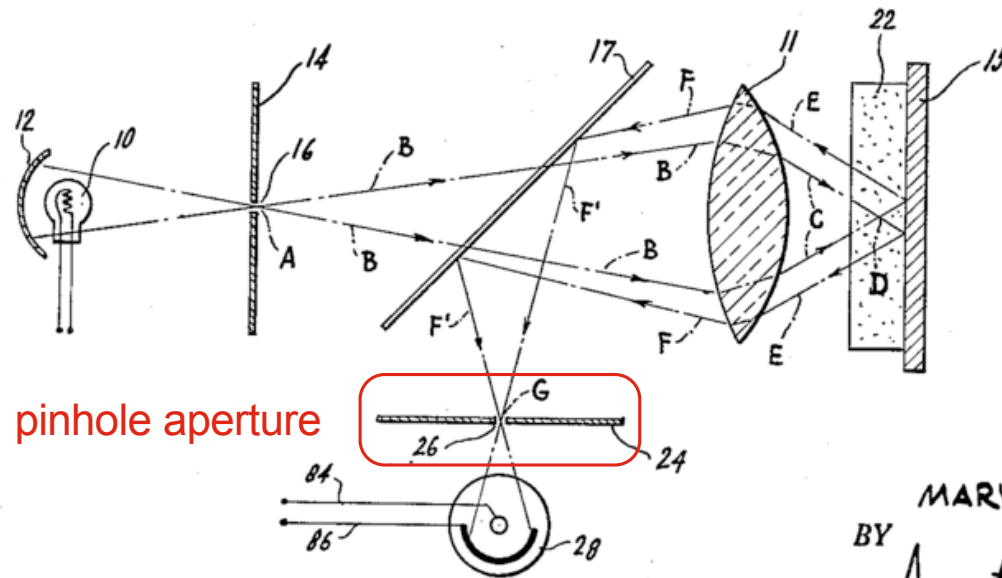


FIG. 3.

pinhole aperture

INVENTOR.

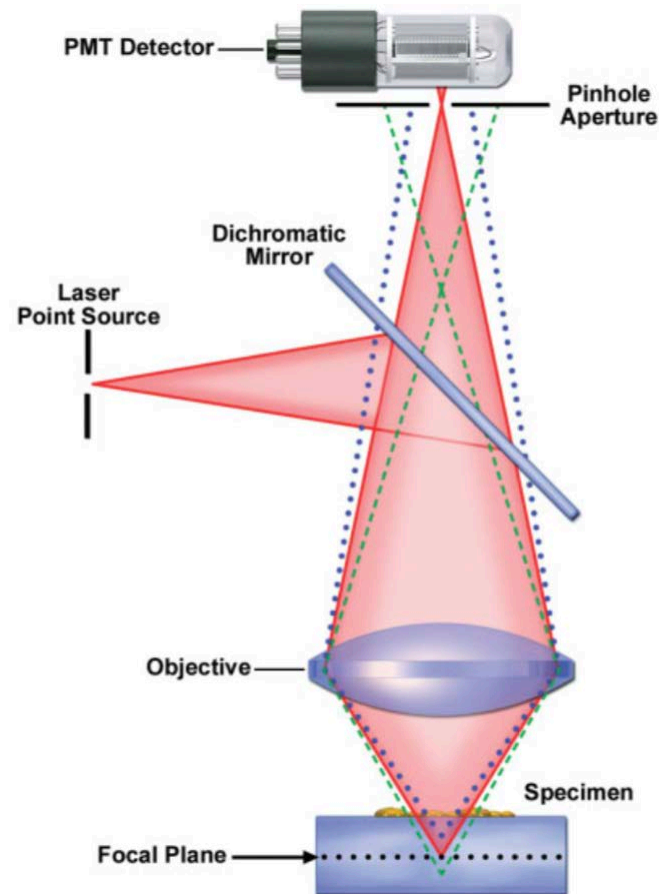
MARVIN MINSKY

BY

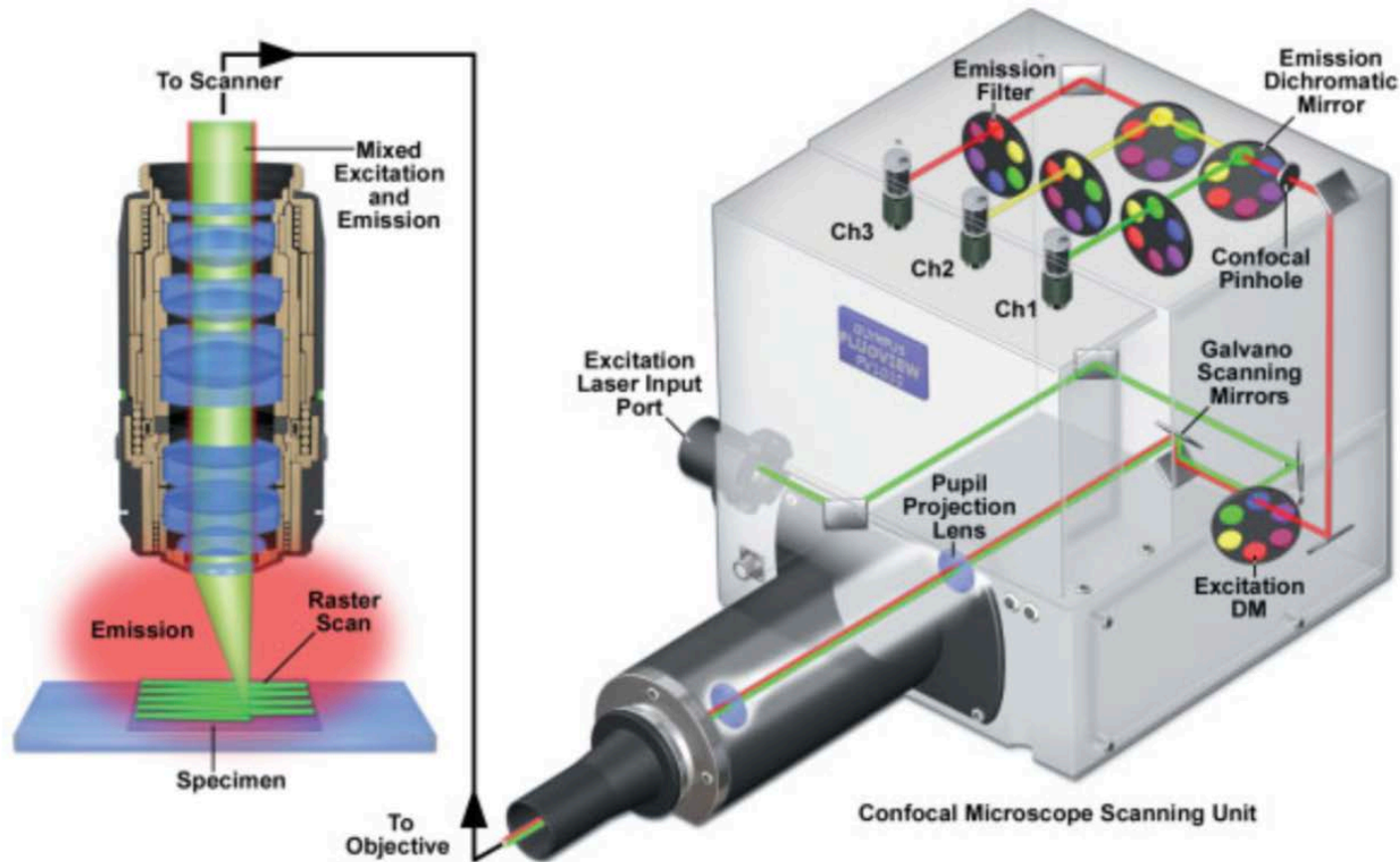
Amster & Levy
ATTORNEYS

Pinhole Aperture: accepts fluorescent photons from the illuminated focused spot, but largely excludes fluorescence signals from objects above and below the focal plane

The confocal principle in epi-fluorescence laser scanning microscopy



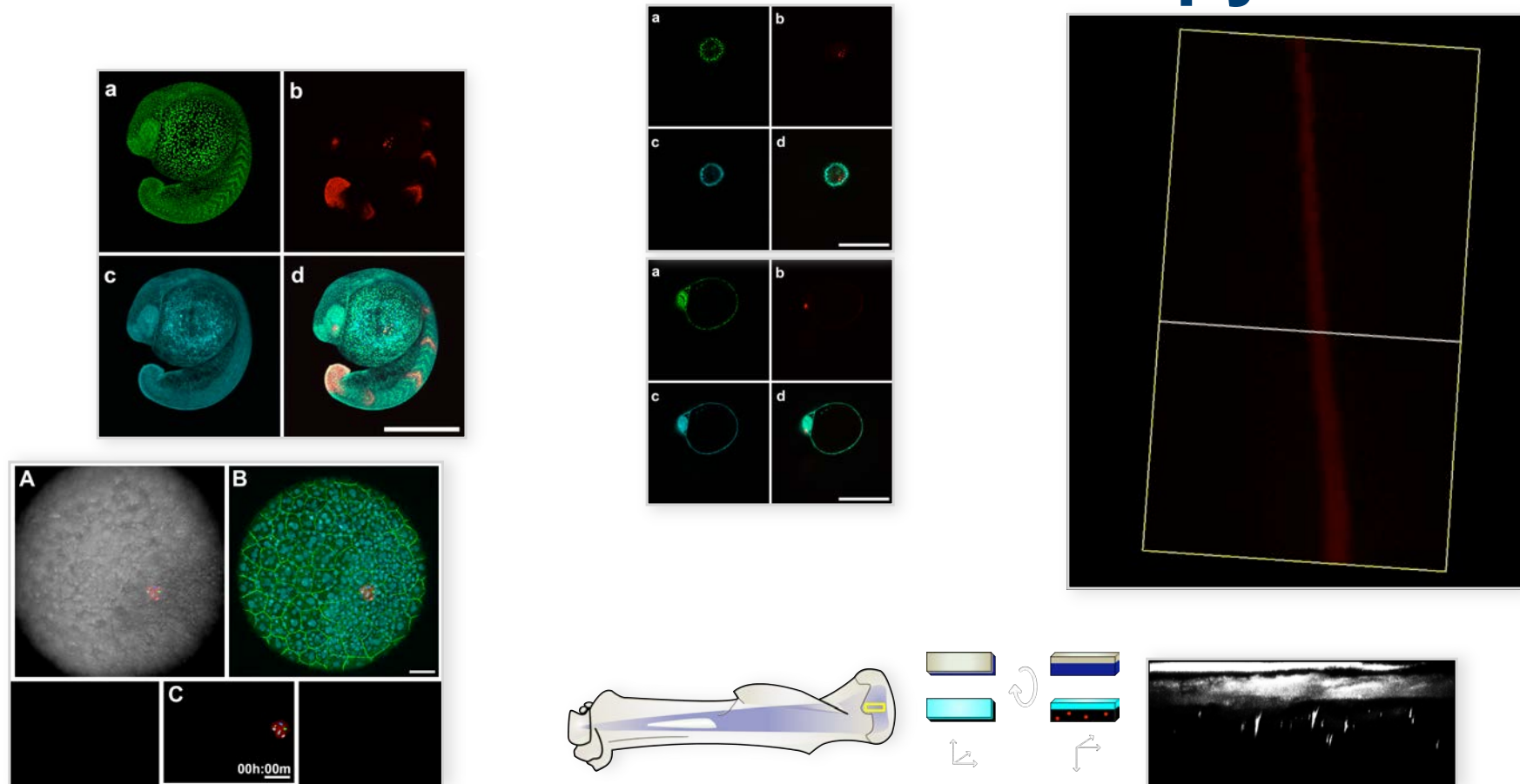
Optical pathway in a confocal microscope point-scanning unit



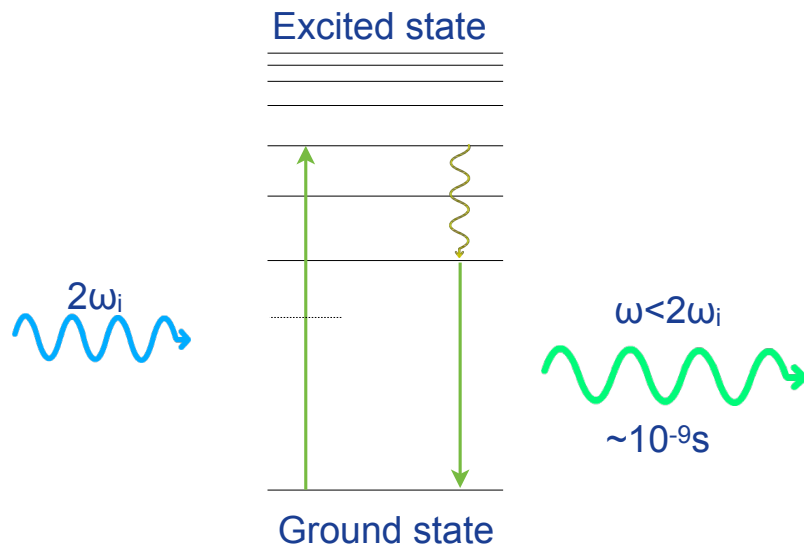
Lasers for fluorophore excitation in CLSM

Laser Type	Wavelength (nm)			
	Violet	Blue	Green	Red
Argon-ion		458, 477, 488	515	
Helium-neon			543	594, 633
Diode	405	440, 473, 488	539, 559	638, 647
Diode-pumped solid state			532, 561	640

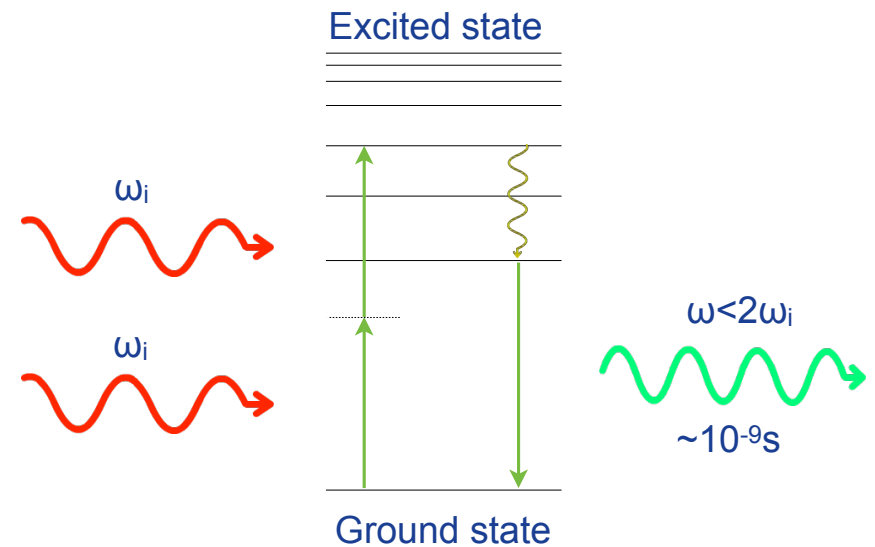
Advantages of confocal over wide-field fluorescence microscopy



1-Photon excited fluorescence versus 2-Photon excited fluorescence

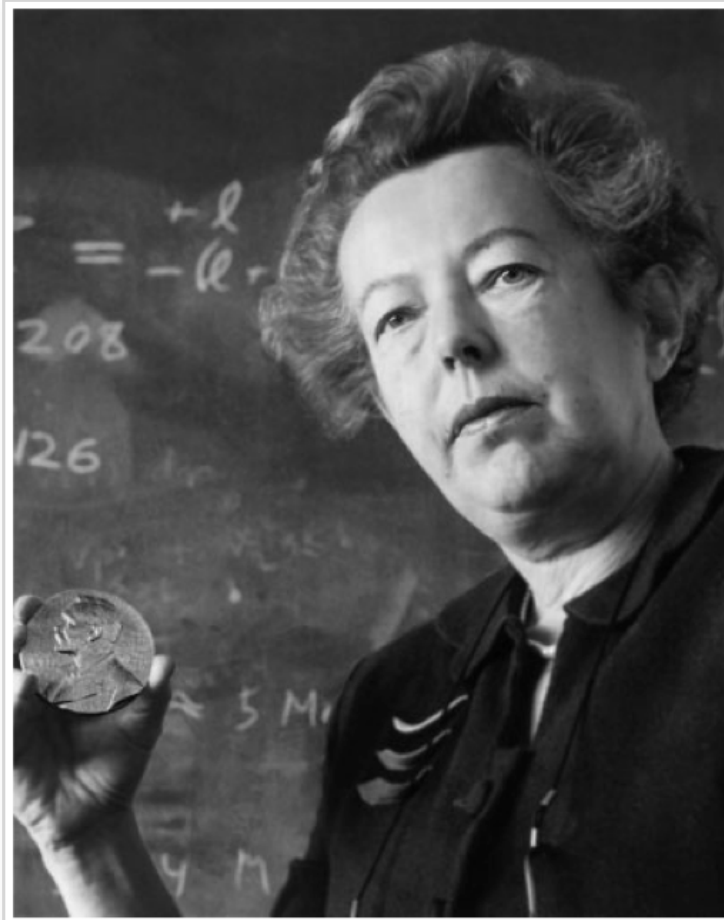


1-Photon excited fluorescence



2-Photon excited fluorescence

Maria Göppert-Mayer (1906–1972)



Probability of molecular excitation

a) *molecular two-photon cross-section, σ*

σ = areas of two excitation photons

x

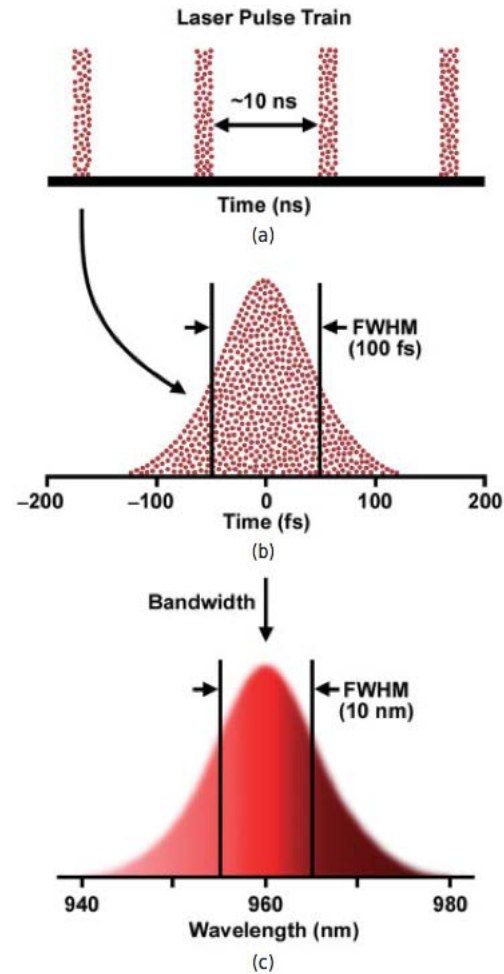
the time within which a target molecule simultaneously absorbs two photons and enters an excited state ($\sim 10^{-18}$ s)

b) *two-photon action cross-sections*

the product of σ and QE, which are given for a particular wavelength in *GM units*

1 GM = $10^{-50} \text{cm}^4 \cdot \text{s} \cdot \text{photon}^{-1}$

The mode-locked Ti:sapphire laser produces pulsed NIR light

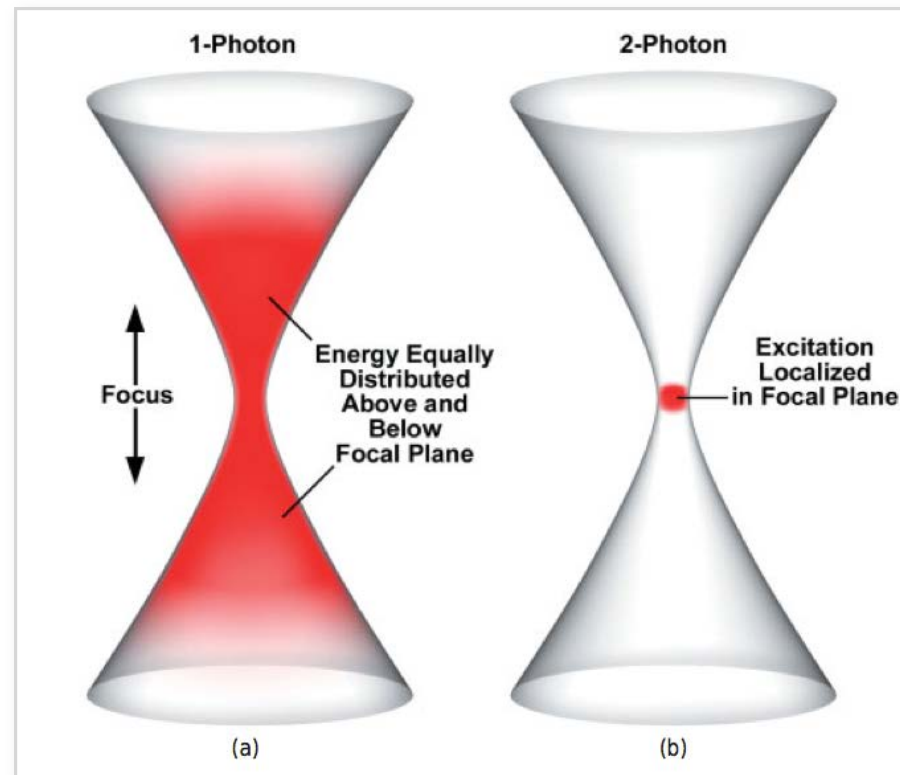


Factors governing total fluorescence present at different distances from the focal plane by one-photon and two-photon mechanisms

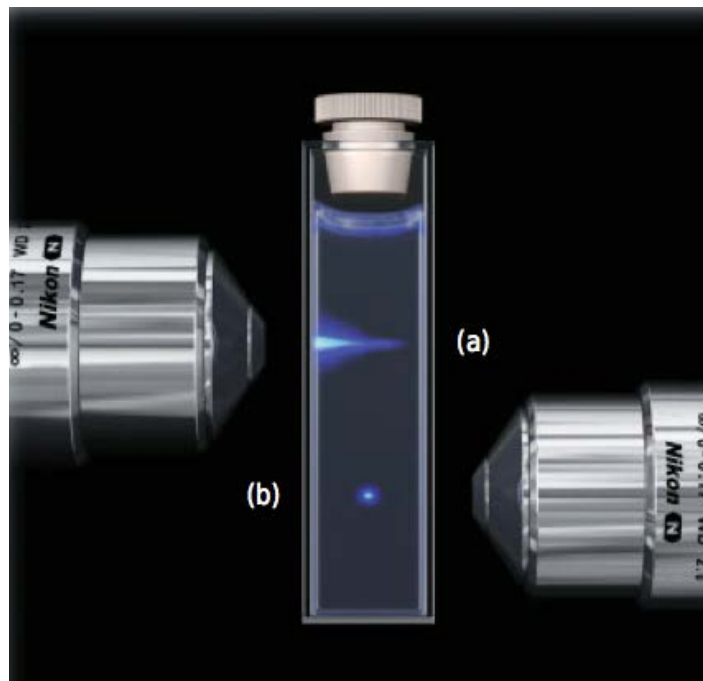
Terms	1 Photon	2 Photon
Absorption rate ϕ	$\phi \sim I$	$\phi \sim I^2$
Intensity I	$I \sim 1/z^2$	$I \sim 1/z^2$
Rewriting ϕ	$\phi \sim I/z^2$	$\phi \sim I/z^4$
Area A of illuminated spot at each distance z from the focal plane ^a	$A \sim z^2$	$A \sim z^2$
Total amount of fluorescence Φ at each focal plane	$\Phi \sim \phi A \sim 1/z^2 \cdot z^2 = 1$ Φ is constant in each cross-sectional area at every distance z from the focal plane	$\Phi \sim \phi A \sim 1/z^4 \cdot z^2 = 1/z^2$ Φ decreases as the square of the z distance, which localizes fluorescence to the focal plane

A key difference between one-photon and two-photon excitation is that the absorption rate (ϕ) is directly proportional to intensity I in one-photon excitation, but is proportional to I^2 in two-photon excitation and is thus a nonlinear process.

Localisation of excitation by one-photon excitation versus two-photon excitation

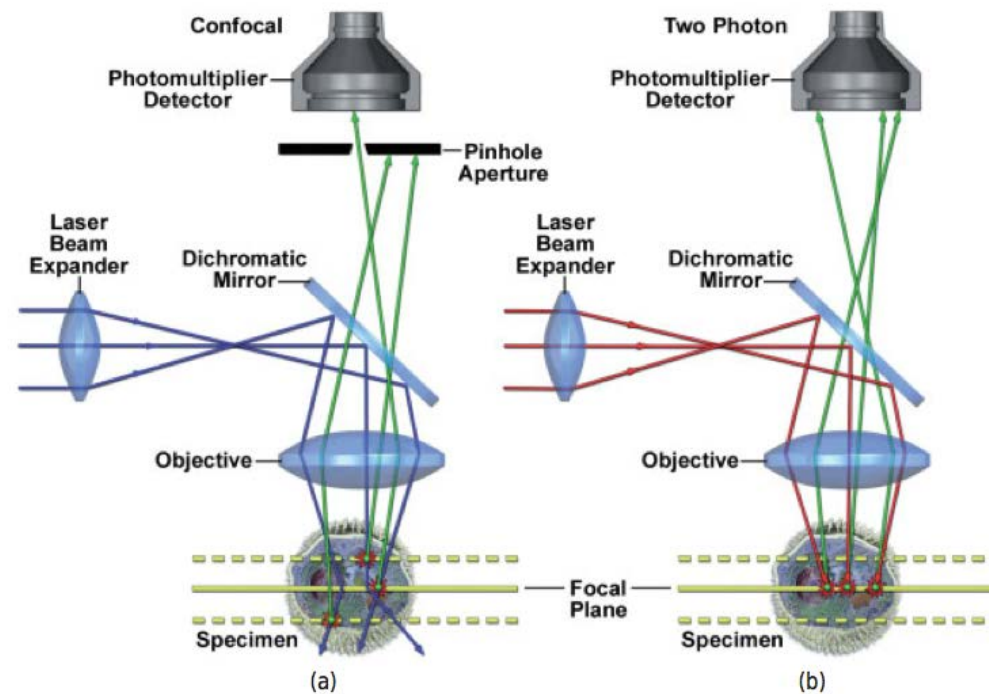


Localisation of excitation by one-photon excitation versus two-photon excitation



Two-photon excitation diminishes with the 4th power of the distance from the focal plane

Confocal imaging versus two-photon imaging



Two-Photon Microscopy:
pulsed near-infrared (NIR) laser & localised excitation of fluorescence

One-photon versus two-photon microscopy

Challenge:

Image cell nuclei/membrane dynamics

a) in cultured cells or b) in cultured tissue explants.

Note:

nuclei are labeled with hoechst 33342 vital stain,
membranes are labeled with membrane-mCherry.

Task:

Please justify which imaging modality you would use for a)
and b).

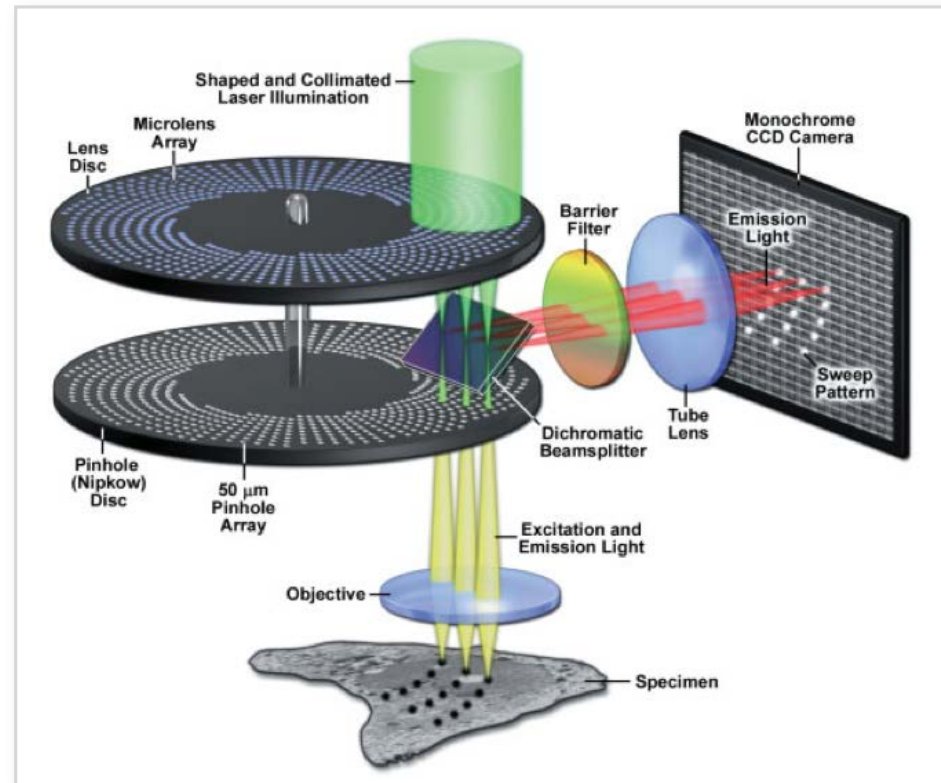
What we will cover

- How to perform fast imaging?
 - How to perform fast volumetric imaging?
 - How to visualise molecular dynamics using fluorescence imaging?
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Confocal Imaging with a Spinning Disk: The Yokogawa design



70% incident light delivered
1-2% fluorescent emission captured

The pinhole spiral patterns are designed so that a complete image is created with each 30° rotation of the disk.

Comparison of spinning disk and single-point scanning systems

Parameter	SD Fast System	SD Standard System	Point Scanning Confocal
Max spin rate	10,000	1,800	N/A
Frame rate	2,000	360	~1-2

Pro/Con of using spinning disk

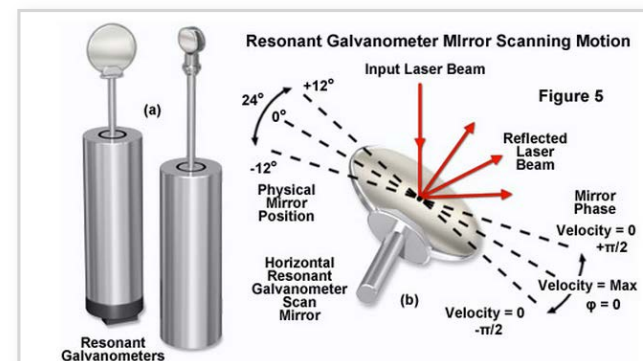
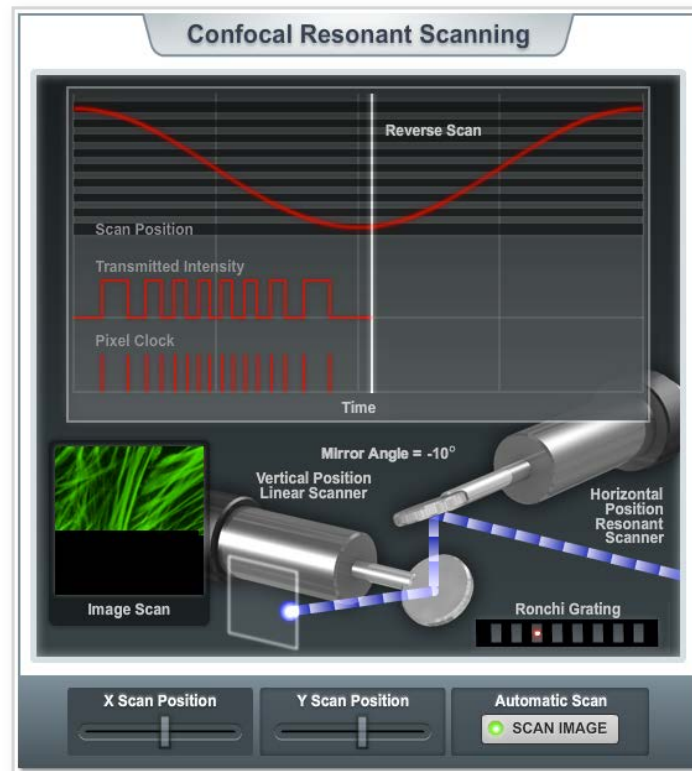
PRO

- The fluorescent light generates an extended real confocal image that can be seen by the eye or recorded on a camera.
- With a sensitive CCD camera as a detector (rather than a PMT), the dynamic range in the image is extended to thousands of gray levels, and light intensity can be quantitated with greater accuracy.
- the total amount of exposure of the specimen is reduced, thus reducing the rate of photobleaching and prolonging cell viability.
- Time resolution is greatly improved. For complete scanning coverage, the fastest spinning disk systems provide up to 2000 full fps versus ~1 full fps for single- point scanning confocals.
- Scanning at a rapid rate but at relatively low intensity results in efficient fluorescence emission.
- Spinning disk systems are also simpler and easier to use than single-point scanning systems.

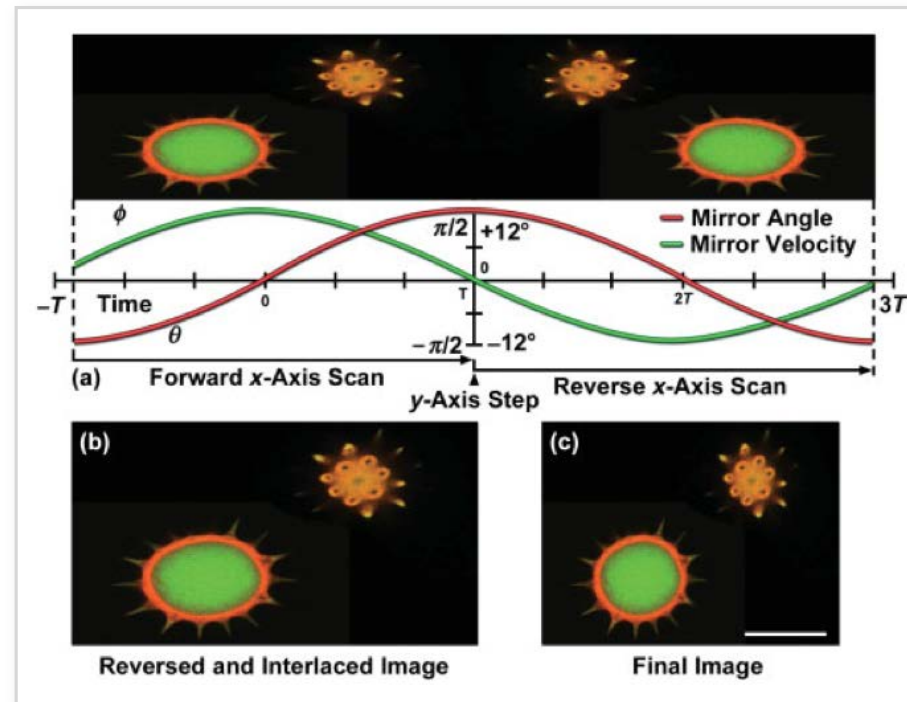
CON

- there is some crosstalk, because the fluorescent light emitted from one site enters several pinholes on the disk, not just one pinhole. This crosstalk adds to the background signal (background haze), reducing the S/N ratio of the object.
- pinholes in the disk are fixed in physical size and not variable. Thus, imaging may be somewhat compromised in resolution and z-section thickness for certain objectives.

Resonant Scanning in Laser Scanning Confocal Microscopy



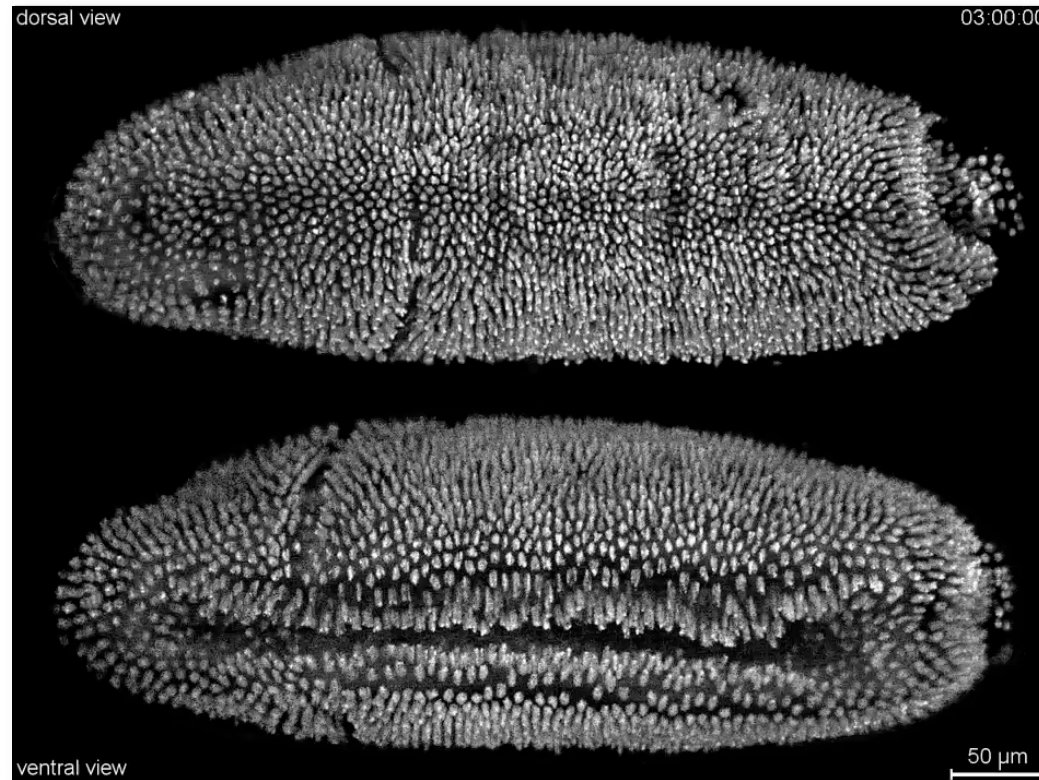
Resonant galvanometers display a nonlinear velocity



What we will cover

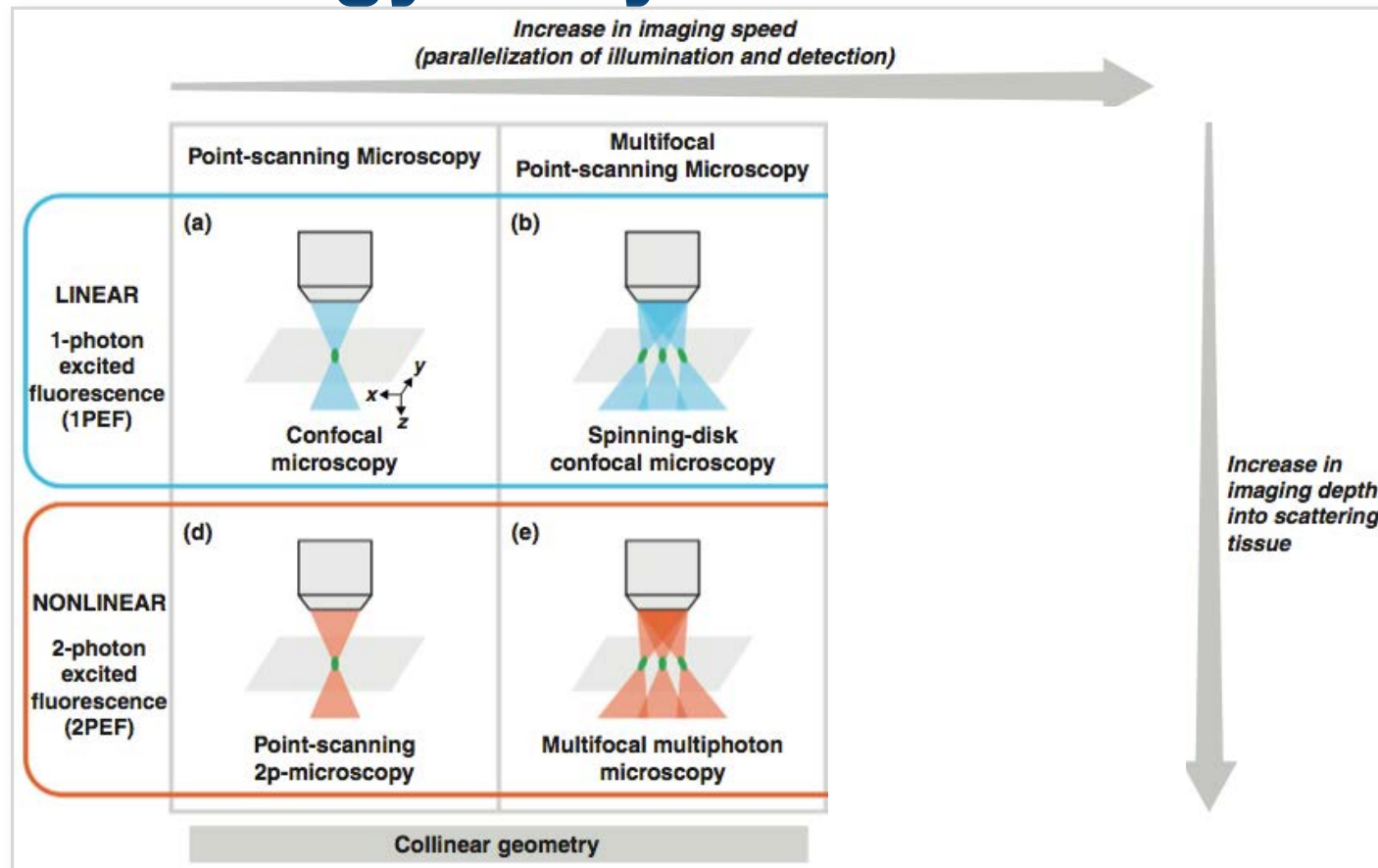
- How to perform fast imaging?
 - **How to perform fast volumetric imaging?**
 - How to visualise molecular dynamics using fluorescence imaging?
-

Why the need to image fast in space? Biology is dynamic and 3D!



Why the need to image fast in space?

Biology is dynamic and 3D!



Long history of light-sheet microscopy

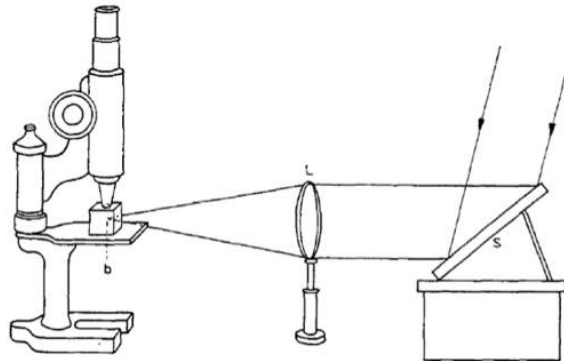


Fig. 1. The first arrangement for making ultramicroscopic particles visible.

Figures from the Chemistry Nobel Lecture 1925

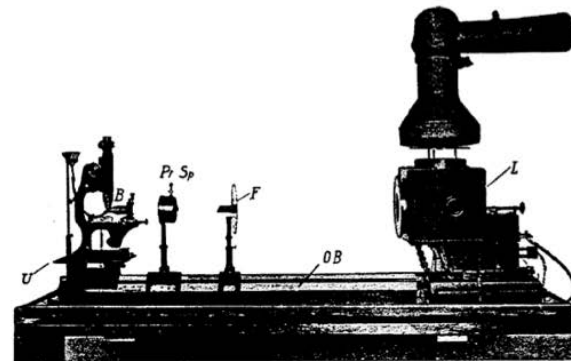


Fig. 2. Immersion ultramicroscope.

*"Ultramicroscope developed by
Siedentopf & Zsigmondy (1902)"*

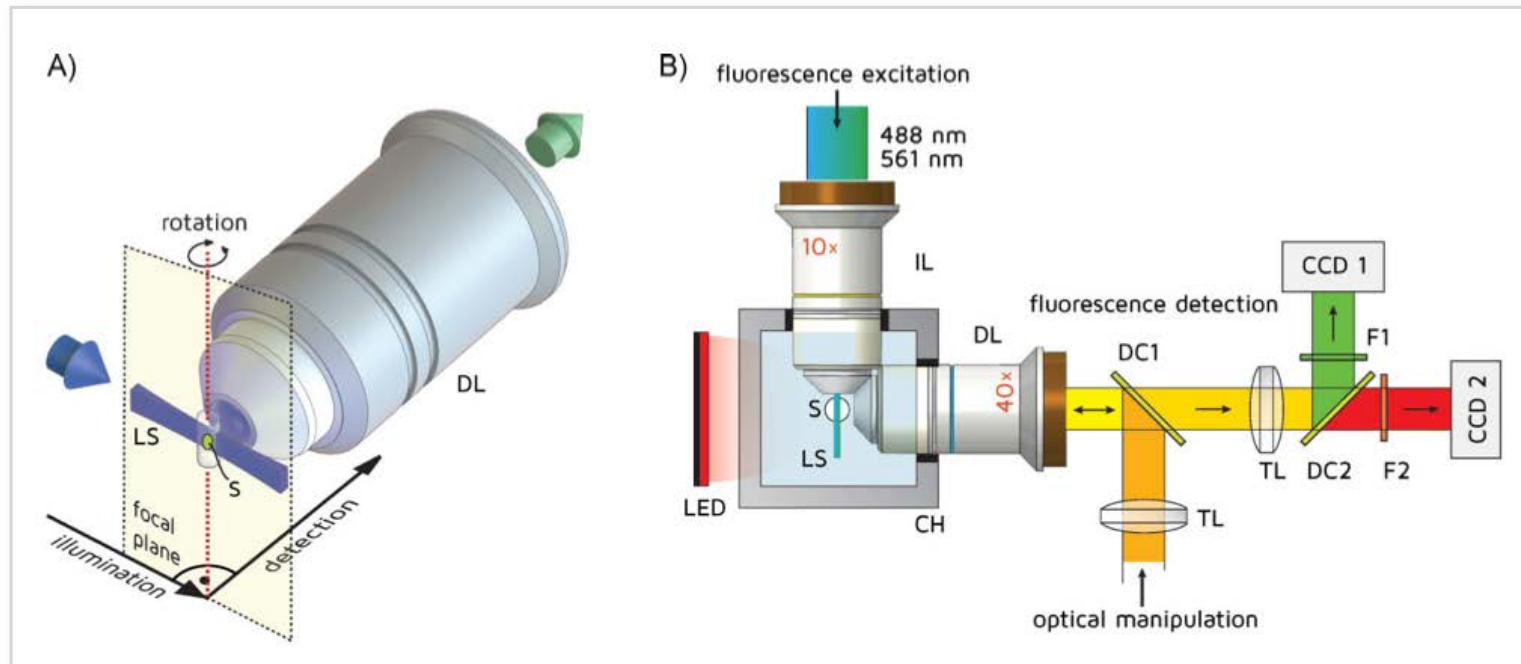


Richard Adolf Zsigmondy

Nobel Prize in Chemistry 1925

**"for his demonstration of the heterogeneous nature
of colloid solutions and for the methods he used,..."**

Selective plane illumination microscopy (SPIM) concept and setup



Only a thin slice of the sample is illuminated by the light sheet and imaged onto the camera;
no out-of-focus blur is created.

Photo-toxicity and bleaching of the fluorophores are reduced to a minimum.

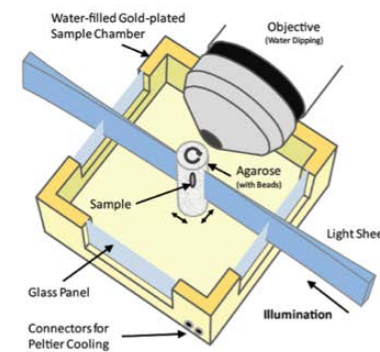
Resolution of light sheet microscopy

Resolution of light sheet microscopy

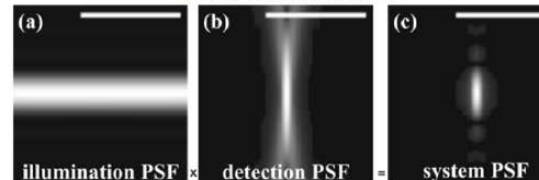
- **Lateral resolution:**
→ detection lens
- **Axial resolution:**
→ illumination and detection lens
- **Field of view:**
→ confocal parameter of illumination

Sheet thickness : $w_0 = \lambda / (\pi \cdot NA_{ill})$

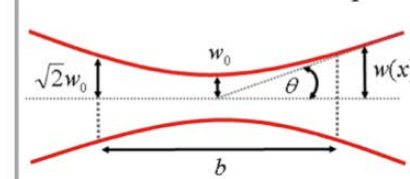
Field of view : $b = 2n\lambda / (\pi \cdot NA_{ill}^2)$



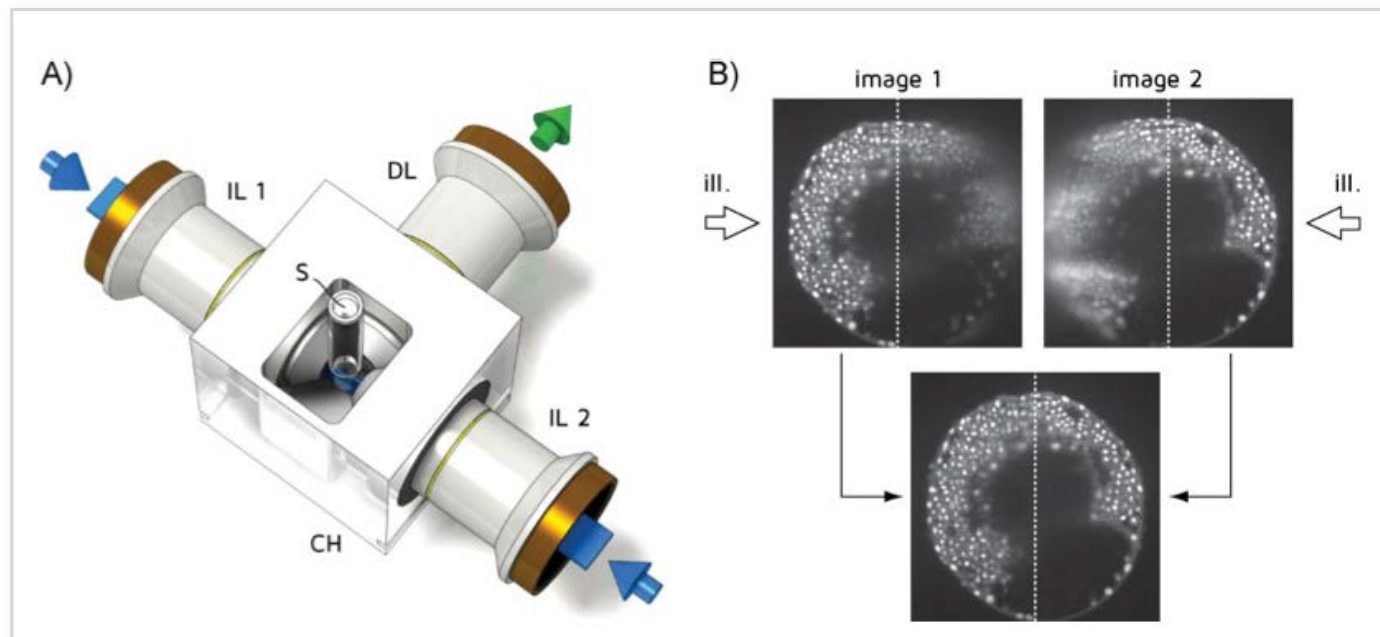
Axial resolution



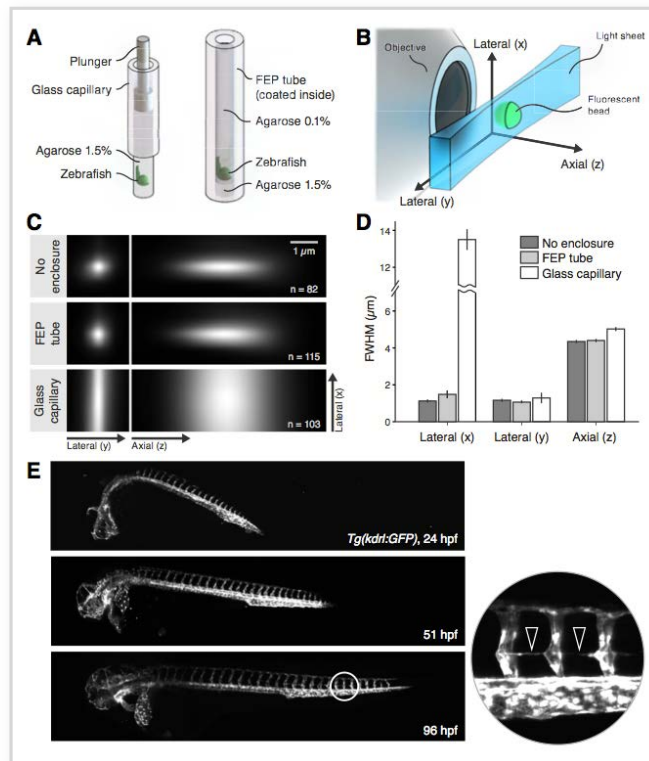
Illumination focus at sample



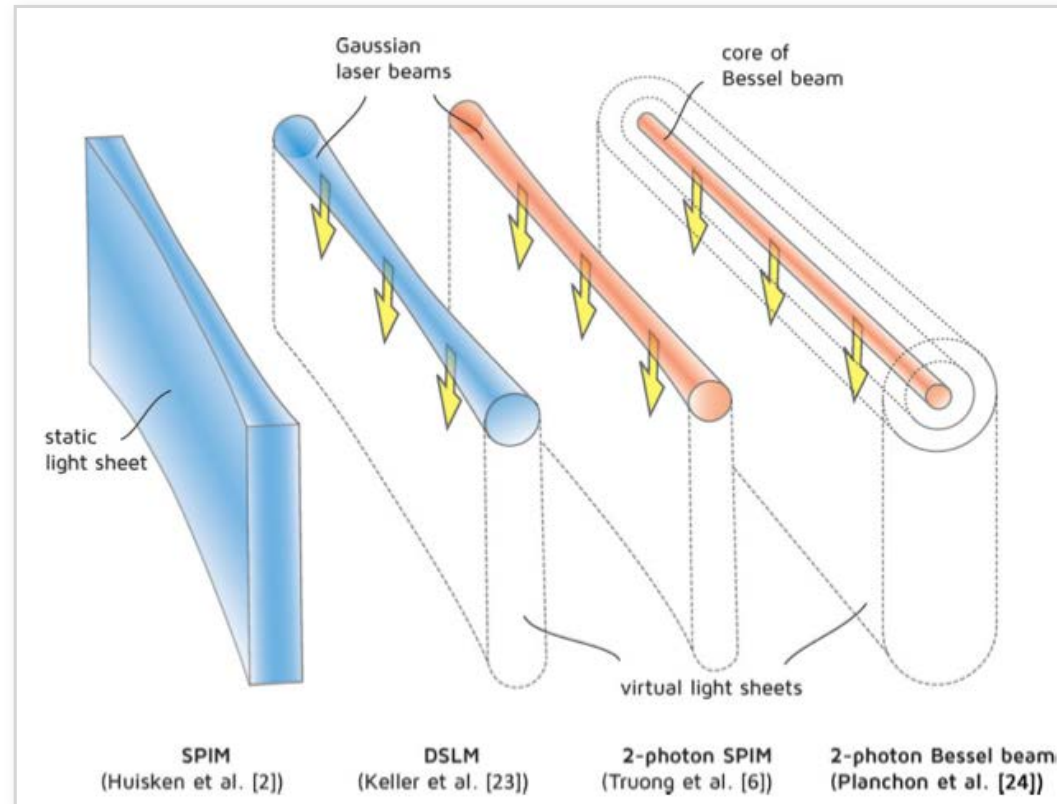
Avoiding imaging artefacts: Dual-sided illumination in multidirectional mSPIM yields an even fluorescence excitation



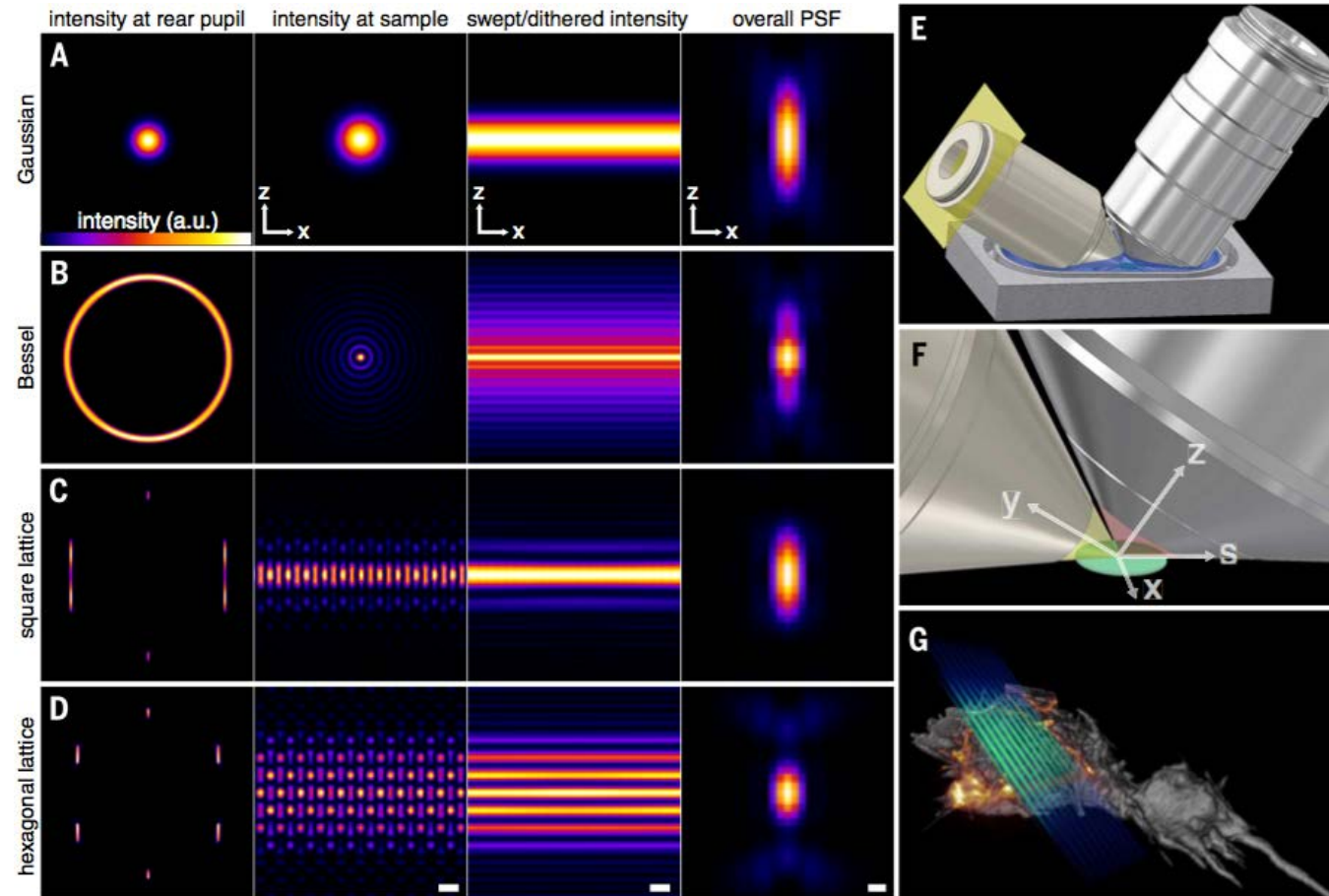
Sample mounting is an issue: improving long-term imaging of zebrafish development in a light sheet microscope



Different forms of light sheets



Lattice light-sheet microscopy

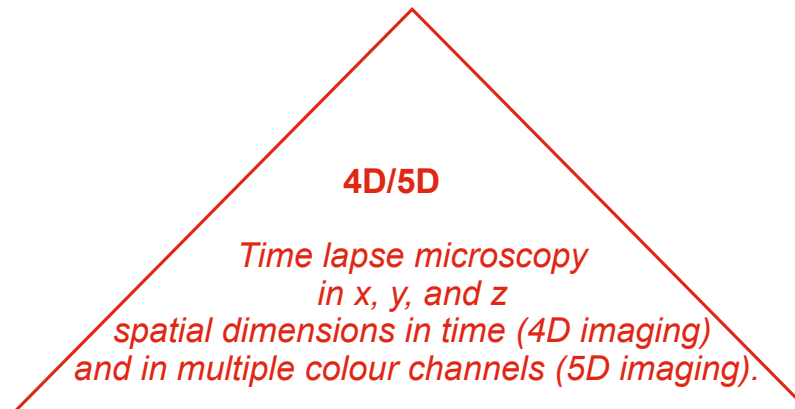


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-

Modes of dynamic fluorescence imaging

FRAP Fluorescence recovery after photobleaching.
FLAP Fluorescence localisation after photobleaching.
FLIP Fluorescence loss in photobleaching.
FDAP Fluorescence decay after photoactivation.



FCS Fluorescence correlation spectroscopy.
FCCS Fluorescence cross-correlation spectroscopy.
ICS Image correlation spectroscopy.

FRET Förster (fluorescence) resonance energy transfer.
BRET Bioluminescence resonance energy transfer.
BIFC Bimolecular Bimolecular fluorescence complementation.
FLIM Fluorescence lifetime imaging microscopy.

FRET:

Förster resonance energy transfer

Zwischenmolekulare Energiewanderung und Fluoreszenz

Von Th. Förster

(Mit 3 Abbildungen)

Inhaltsübersicht

In Weiterentwicklung früherer Theorien von J. und F. Perrin und klassisch-physikalischer Überlegungen des Verfassers wird eine quantenmechanische Behandlung des Übergangs von Elektronenanregungsenergie zwischen gleichartigen Molekülen in Lösung gegeben. Der kritische Molekülabstand, unterhalb dessen der Übergang während der Anregungsdauer stattfindet, läßt sich aus den Absorptions- und Fluoreszenzspektren und der Anregungsdauer der Moleküle berechnen. Für Fluorescein und Chlorophyll *a* ergeben sich Werte von 50 bzw. 80 Å, entsprechend den mittleren Molekülabständen in Lösungen von $3.2 \cdot 10^{-3}$ bzw. $7.7 \cdot 10^{-4}$ Molen/Liter. Für die Bereiche oberhalb und unterhalb der kritischen Konzentration werden Formeln zur Berechnung der Energieabwanderung vom Primärmolekül angegeben, die mit den vorliegenden Messungen der Konzentrationsdepolarisation der Fluoreszenz gut übereinstimmen. Die Anwendung auf analoge Energiewanderungsprobleme in Molekulkristallen und im Assimilationsapparat der Pflanze wird diskutiert.

§ 1. Beobachtungen über Energiewanderung

Es ist bekannt, daß optisch angeregte Atome und Moleküle fähig sind, ihre Elektronenanregungsenergie beim Zusammenstoß mit anderen Atomen oder Molekülen an diese abzugeben¹⁾. In vielen Fällen scheint dies unter Übergang eines Elektrons zu geschehen, so bei der Löschung der Fluoreszenz von Farbstofflösungen durch Fremdstoffe nach einem von Weiß²⁾ angegebenen Mechanismus. In anderen Fällen findet jedoch ein Übergang der Energie ohne Bindung an Materie irgendwelcher Art statt, wie z. B. bei der sensibilisierten Fluoreszenz im Gasraum [Franck und Cario³⁾ ⁴⁾].

Bei großer Dichte der anregbaren Materie, z. B. in absorbierenden Kristallen, wird vielfach eine Wanderung der Anregungsenergie über viele Moleküle hinweg und auf entsprechend große Entfernungen beobachtet. In Kristallen mit Elektronenleitvermögen ist wohl eine Energieübertragung durch wan-

¹⁾ Es ist dies ein Sonderfall der sogenannten Stöße zweiter Art.

²⁾ J. Weiß, *Nature* **141**, 248 (1938).

³⁾ J. Weiß, *Trans. Faraday Soc.* **85**, 48 (1939).

⁴⁾ J. Franck, *Z. Physik* **9**, 859 (1922).

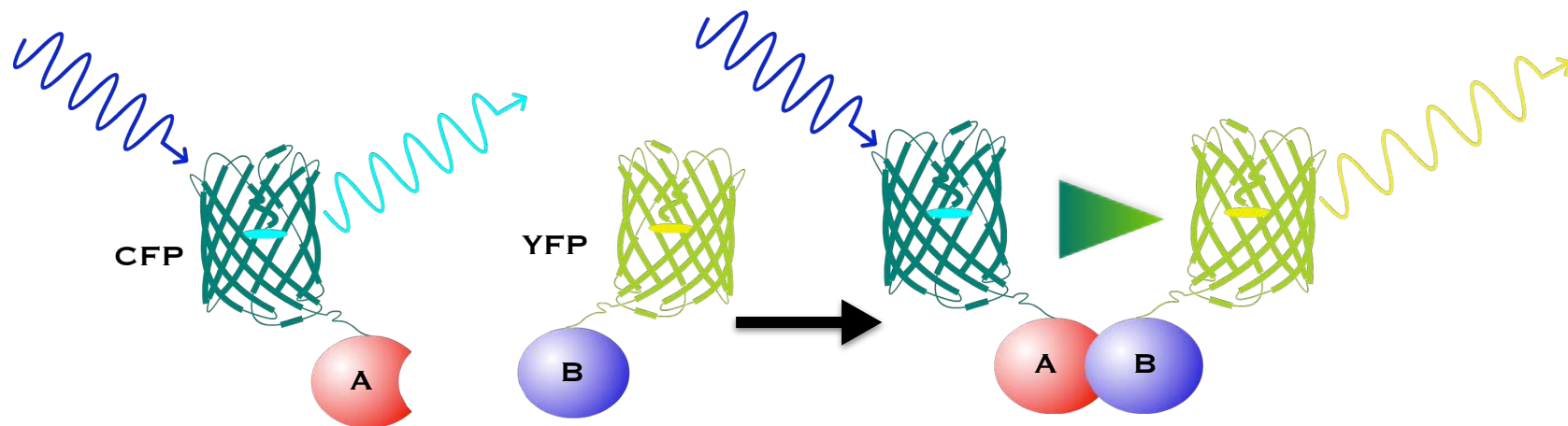
⁵⁾ G. Cario, *Z. Physik* **10**, 185 (1922).

⁶⁾ G. Cario und J. Franck, *Z. Physik* **17**, 202 (1923).



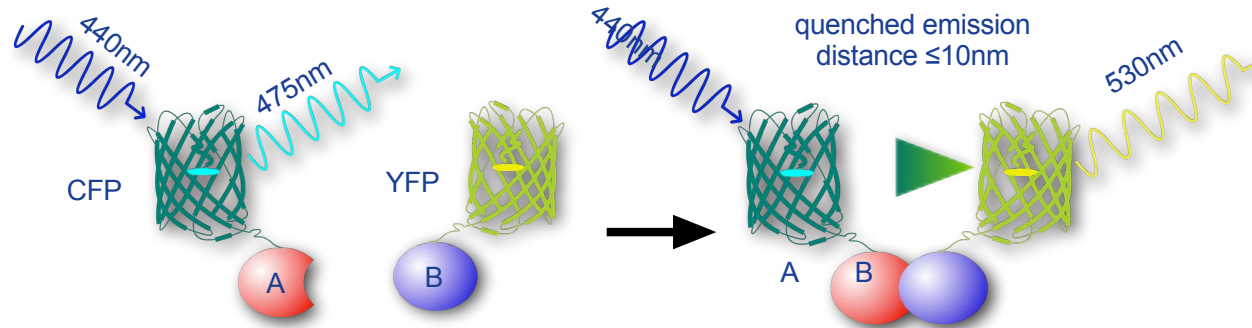
Theodor Förster

Molecular mechanism of FRET



FRET involves the radiation-less transfer of energy between closely spaced donor and acceptor fluorophores. FRET occurs in the absence of emission and absorption of an intermediate photon.

Main requirements for FRET



- **Fluorophore proximity (up to 10nm)**

(Förster distance R_0 = 50% FRET transfer probability)

- **Dipole alignment**

(is greatest when both dipoles are in line and parallel and least when they are perpendicular to one another)

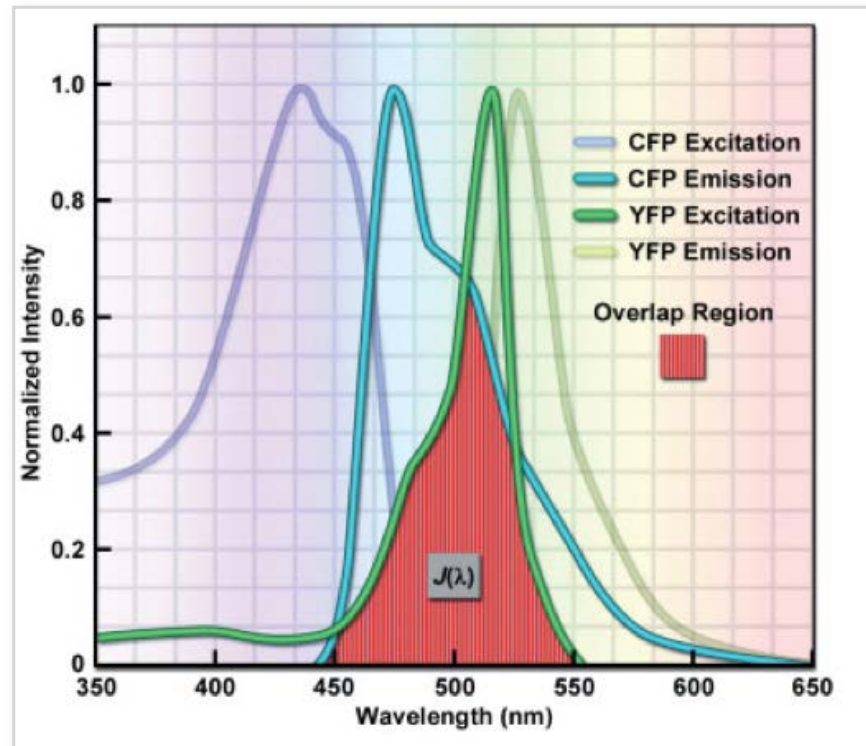
- **Spectral overlap**

(emission spectrum of the donor fluorophore overlap the excitation spectrum of the acceptor fluorophore)

- **Environmental conditions**

(affect dipole interactions, including solvent type, degree of hydration, ionic strength, pH, and temperature)

Main requirements for FRET



Förster radius

$$R_0 = [8.8 \times 10^{23} \cdot \kappa^2 \cdot n^{-4} \cdot QY_D \cdot J(\lambda)]^{1/6} \text{ \AA}$$

where κ^2 = dipole orientation factor (range 0 to 4; $\kappa^2 = 2/3$ for randomly oriented donors and acceptors)

QY_D = fluorescence quantum yield of the donor in the absence of the acceptor

n = refractive index

$J(\lambda)$ = spectral overlap integral (see figure)

$$= \int \epsilon_A(\lambda) \cdot F_D(\lambda) \cdot \lambda^4 d\lambda$$

where ϵ_A = extinction coefficient of acceptor

F_D = fluorescence emission intensity of donor as a fraction of the total integrated intensity

Donor	Acceptor	R_0 (Å)
Fluorescein	Tetramethylrhodamin	55
Cy3	Cy5	>50
CFP	YFP	50
BFP	GFP	40
CFP	GFP	48
GFP	YFP	57

Evaluating FRET from fluorescent images

one can calculate the FRET efficiency, E , and the intermolecular distance, R .

E is calculated from the ratio of fluorescence intensities of the donor with acceptor (I_{da}) and without the acceptor (I_d):

$$E = 1 - (I_{da} / I_d).$$

FRET efficiency is related to the intermolecular distance R between the donor and acceptor fluorophores.

Since E and R are related:

$$E = R_0^6 / (R_0^6 + R^6)$$

dipole–dipole coupling and FRET efficiency are related to the inverse 6th power of the donor–acceptor distance. This degree of sensitivity makes FRET very useful as a “spectroscopic ruler” for measuring intermolecular distances in the range of 1–10 nm.

Paradigms for FRET

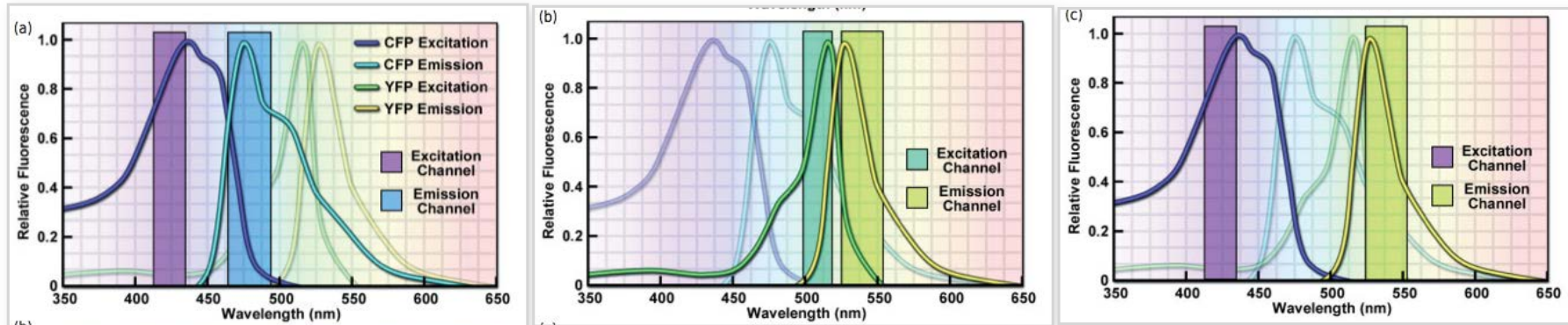
- **Intermolecular FRET:**

Donor–acceptor fluorophores are located on two separate molecules that interact with each other, and FRET indicates the proximity of the two reporters, and indirectly, the proximity of the two test molecules.

- **Intramolecular FRET:**

Donor-acceptor fluorophores are placed on different regions of a single polypeptide that undergoes conformational changes upon binding to a substrate, causing the fluorophores involved in FRET to come together or move apart.

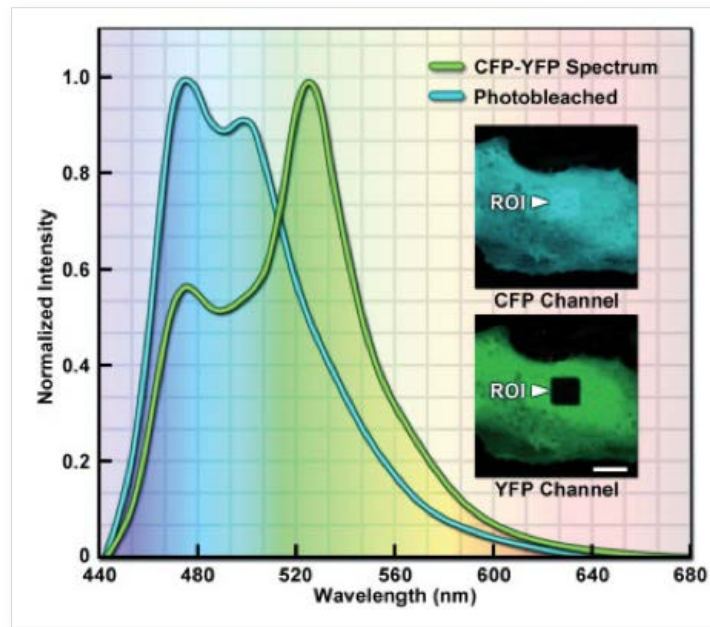
Methods for detecting FRET: Sensitised Acceptor Emission FRET



The operator must calculate correction factors for:

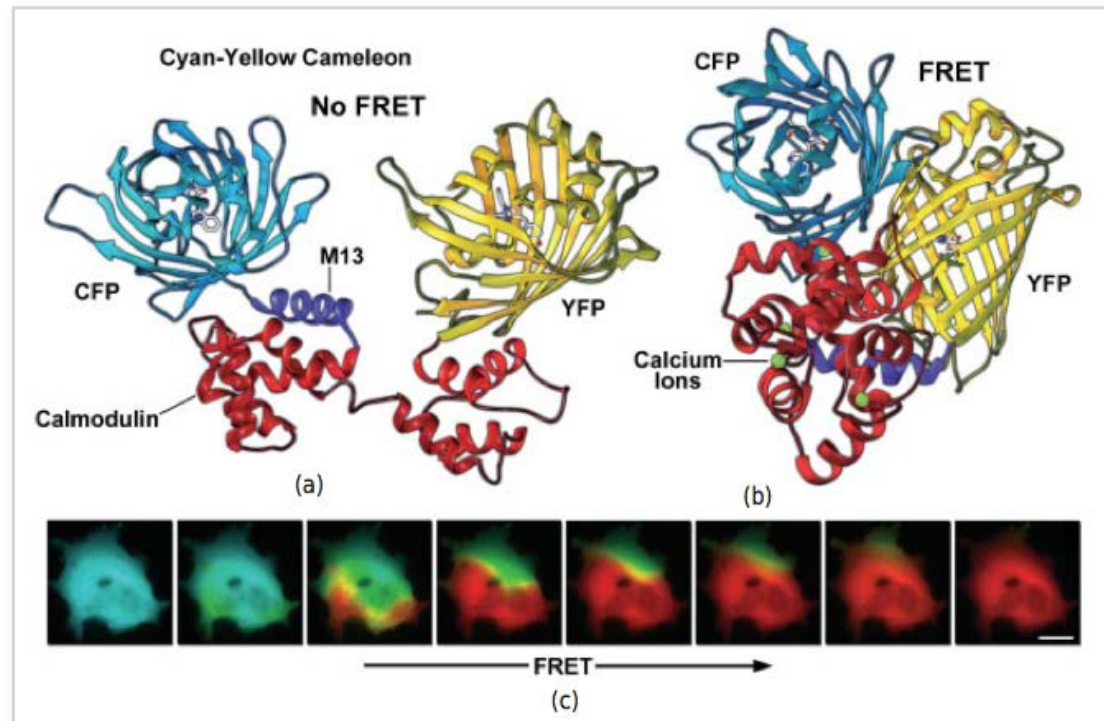
- **donor bleedthrough** (donor fluorescence that contributes to signal in the FRET image)
- **crosstalk** (donor excitation that directly excites acceptor fluorescence).

Methods for detecting FRET: Acceptor-Photobleaching FRET with de-quenching of donor fluorescence



The operator must ensure that the illumination used to bleach acceptor fluorescence does not also destroy some of the donor fluorescence.

Applications for FRET: Cameleons-biosensors for Ca^{2+} ions



FRET:

additional considerations

- The concentrations of donor and acceptor need to be controlled so that the efficiency of FRET transfer is optimal. Ideally, there is molar excess of acceptor over donor (but not a 10-fold or greater molar excess).
- Photobleaching should be minimal to maximise the number of functional molecules generating the FRET signal.
- Spectra for donor emission and acceptor excitation must overlap substantially.
- The best FRET pairs have a donor with a large fluorescence quantum yield and an acceptor with a large absorption extinction coefficient.
- Minimise direct excitation of acceptor during excitation of donor.
- Donor's absorption and emission spectra should overlap minimally to avoid donor–donor transfer, the case where emission of the donor molecules stimulates other donors to fluoresce, thus increasing a nonspecific background signal.
- Labeled donor and acceptor molecules must retain their biological functions.
- Direct excitation of the acceptor at the donor excitation wavelength. This is unwanted but is usually unavoidable, since the excitation spectra overlap and have long tails.
- Possibility of Emission of the excited donor at the acceptor emission wavelengths.
- Effects of varying concentrations of donor and acceptor molecules.
- Poor filter performance and therefore poor spectral discrimination when performing FRET.

Thank you!

Periklis (Laki) Pantazis
